Deadline-aware inference batching

Worked example · synthetic data

A page-reading service receives requests in bursts. Larger batches can improve throughput, but waiting for a batch also consumes the response-time budget. This worked example compares fixed batching with a deadline-aware rule. All measurements below are illustrative, not benchmark results.

1. Workload and queueing estimate

The synthetic workload contains short text pages, equation-heavy notes and pages with tables. The average arrival rate is 18 requests per second, with brief bursts above that average. Requests are served in arrival order; the response-time target is a 95th percentile below 120 ms.

For a single-worker approximation, let the mean service time be 21 ms and the second moment be 0.00061 s². The utilization and mean queueing delay are:

$$\rho = \lambda\,\mathbb{E}[S] = 0.378,$$

$$\mathbb{E}[W_q] = \frac{\lambda\,\mathbb{E}[S^2]}{2(1-\rho)} \approx 8.8\,\text{ms}.$$

This estimate assumes independent arrivals and a stable queue. It predicts a mean, not a tail-latency guarantee; bursty traffic must still be checked separately.

2. Deadline-aware rule

Let q_t be the number of queued pages. Choose a batch cap using the rule below, then dispatch when either that cap is reached or the oldest page has waited 8 ms. A partially filled batch is allowed. This avoids holding a quiet queue until eight pages arrive.

$$b_t = \min\left(8, \max\left(1, \lceil q_t / 2 \rceil\right)\right).$$

3. Illustrative comparison

Policy	Batch cap	Throughput (req/s)	p95 (ms)
Single page	1	24	62
Fixed batch	4	58	118
Fixed batch	8	76	204
Deadline-aware	8	63	96

Fixed batches of eight have the highest throughput in this example but miss the 120 ms target. The deadline-aware policy gives up some throughput and keeps the illustrative p95 at 96 ms. These values describe the example workload only; they do not compare AI models or hardware.

4. Reading the result

Read the equations, units and table together. A better mean does not imply a better 95th percentile, and a batch cap is not the number of pages in every dispatch. Record queue depth and the oldest waiting time before interpreting an apparent speed-up.

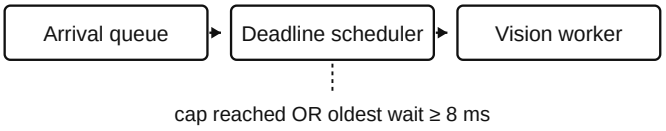


Figure 1. The scheduler releases a batch when its size limit or waiting-time limit is reached.

1. p95 is the 95th-percentile response time. Synthetic values are provided for a reproducible reading example, not a performance claim.